WHEN PERFECT COARSE PREDICTIONS ARE NOT ENOUGH FOR ACTION EVALUATION

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ABSTRACT

Does accurately predicting a spatial supervision target validate the action measurement computed from it? We study a procedural visual task with two attribute fields, four candidate footprints, and a public exposure-to-cost law. A common-reference oracle decomposition separates supervision-fit error, source-raster difference, and the error of interpreting coarse attributes as uniform within cells. The last term is an elementary within-cell covariance, but its consequences for actual action acceptance require measurement. Discovery motivates one frozen 2,000-scene IID confirmation: the separately pooled oracle loses 63 safe opportunities, with 61 remaining under a finer-property check; perfect prediction of the actual supervision target still loses 54. At fixed model-selected actions, pooling changes 3.22% of danger labels versus 0.15% for the numerical check. Observed changes are conservative and concentrate at overlapping boundaries with small cost margins. We then append one same-information comparison of uniform interpretation, a development-selected global shift, and an established area-preserving interface reconstruction. On the consumed bank, reconstruction recovers 44 of the target oracle's 54 lost opportunities but adds 16 unsafe accepts; the frozen global shift recovers six and adds three. Thus the current target–aggregation interface is a bottleneck, without proving an impossibility of coarse representations or a risk-free repair. The study distinguishes fitting a learning target from validating its downstream measurement, without proposing a new identity, reconstruction method, or safety certificate. Evidence is restricted to the stated synthetic task.

1 INTRODUCTION

Spatial predictions often enter a decision through a second computation: a map is integrated over a proposed action footprint and converted to a cost. Improving the predictor is useful only in relation to the quantity this computation should measure. If the target itself is a coarse spatial average, an accurate prediction can still be interpreted incorrectly inside each cell. A failure after perception therefore does not identify whether learning, spatial interpretation, or the decision rule is the appropriate object to change.

Our motivating pipeline learned two visual attribute fields and evaluated four supplied actions with a known cost formula. Geometry repair improved ranking, but a joint residual bound still rejected every action. Earlier diagnostics separated output postprocessing from perception weights and showed that a simple fixed rule could pass an existing selective-risk screen. They did not explain what the spatial target preserved for action measurement. A low-resolution versus reference comparison was insufficient: it changed both attribute and footprint rasters, conflating the source of the difference.

We address a narrower question: *how can one distinguish failure to realize a supervision target from failure of the target's current aggregation to realize the intended task measurement?* We replace predictions by the exact supervision target, then by a coarse target derived from the same high-resolution reference, while keeping the action footprint fixed. Each replacement answers a different question. Perfect target prediction removes supervision-fit error but does not change the aggregation rule. Common-source pooling removes the source-raster difference but still assumes uniform within-cell occupancy. Direct integration retains the spatial coincidence between an attribute and the action.

The resulting decision effect is primarily conservative in this task. In one prospective IID confirmation, the coarse oracle discards safe opportunities even after choosing its own action. On the learned model's fixed choices, all observed pooling label changes are reference-safe to pool-danger. The finding concerns lost usable actions, not an observed rise in unsafe acceptance from this term. Signed learning and measurement errors can partly cancel, so a near-zero mean error cannot establish that either component is accurate.

Crucially, failure of a uniform-cell aggregator does not prove failure of the coarse representation for every decoder. Neighboring fractions can retain boundary information. We therefore add one established piecewise-linear interface reconstruction (PLIC) and a simple global shift, with the same coarse fields and action information. The comparison is explicitly retrospective on an already consumed bank; it does not replace the prior confirmation or authorize a third one. Its purpose is to determine the strength of the interpretation.

Our contributions are (i) a common-reference oracle attribution separating learning, source-raster, and aggregation terms; (ii) measured decision effects and a prospective check of their direction and operating conditions in the normal generator; and (iii) one same-information control that tests whether changing spatial interpretation is more appropriate than treating the residual error as evidence of insufficient model capacity. The covariance identity, the general prediction–decision distinction, and PLIC are established ideas. The contribution is their use to identify and delimit an actual decision bottleneck.

2 Task, learning target, and decision objects

Visual task. A top-down $64{\times}64$ RGB image contains two disjoint patches assigned water and fragile-surface attributes. Water has wave texture and fragile regions have crack texture; appearance families vary color, background, and illumination. Each continuous scene has two rendered versions with exchanged attributes but identical shapes and paths (Figure 1). The normal generator samples freely positioned and rotated ellipses or boxes subject to fixed continuous in-frame and nonoverlap constraints. It does not reject scenes based on raster labels, cost, or candidate safety. This is a controlled visual task, not a natural-image or physical-hazard benchmark.

Four polylines share start and goal points on opposite scene boundaries; each has two independently drawn interior waypoints. A swept radius of .05–.10 in normalized coordinates defines each footprint as a spatial union. Repeated traversal is not counted repeatedly. Candidate construction is independent of the attribute assignment and need not supply a safe action. All comparisons retain the same four paths, card, image version, tie variable, and danger threshold. No planner or new candidate is introduced.

Learning setup. The predictor is a randomly initialized ResNet-18 (He et al., 2016) with four lateral $1{\times}1$ projections to 64 channels, bilinear top-down fusion, $3{\times}3$ refinement, and a two-channel output at $16{\times}16$. Sigmoid outputs fit area-pooled binary attribute masks using binary cross entropy plus soft Dice loss. Each fixed model uses 800 training images, one version per scene, 24 epochs, batch size 20, and 960 AdamW updates (learning rate .001; weight decay .0001), with the final checkpoint rather than validation-based selection. Color/polarity/contrast augmentation preserves geometry. There are no high-resolution action-cost labels or candidate-conditioned inputs to this training. We retain all five previously fixed independent-geometry/source 64 checkpoints.

Public cost and actual deployment unit. Mean footprint exposures are $e = (e_w, e_f)$. Cards A, C, and D define

$$g_A(e) = e_w, \qquad g_C(e) = e_w + e_f - e_w e_f, \qquad g_D(e) = e_f. \tag{1}$$

The trivial zero-cost card B is excluded from deployment. Each independent scene draws one of the two image versions and one A/C/D card uniformly. The selector minimizes predicted cost; ties within 10^{-9} use a stored independent uniform draw. It chooses once, then accepts iff the estimate is at most $\tau = .02$. Rejection never triggers reselection. The direct 512 attribute–footprint integral and Eq. 1 define reference cost. It is a synthetic mean-exposure criterion, not collision probability, traction, or execution success.

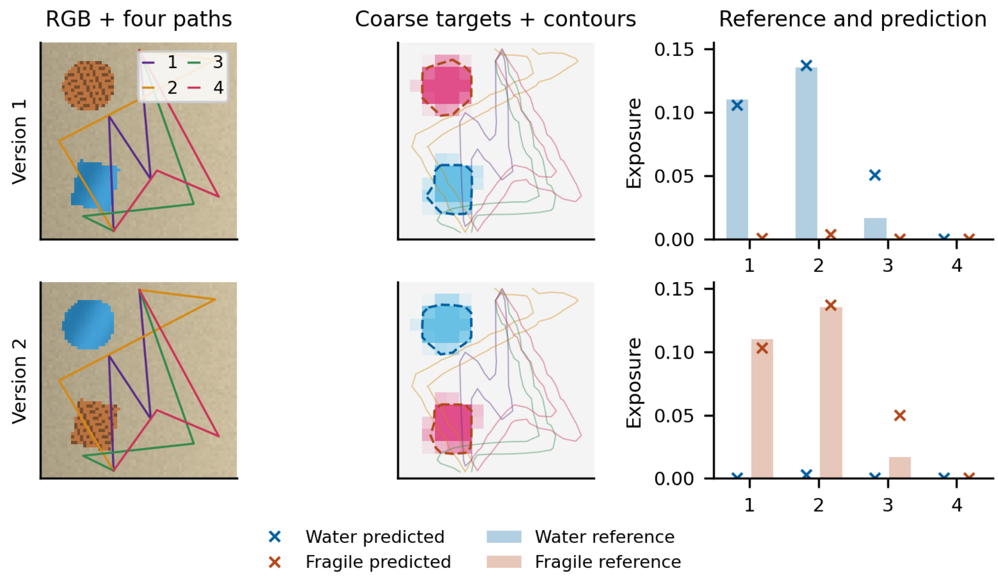

Figure 1: Fixed first discovery scene and its exchanged-attribute version. Left: RGB and candidate centerlines. Middle: source 64-to 16 targets (cyan: water; magenta: fragile), footprint contours, and dashed predicted .5 contours. Right: direct 512 and stored predicted exposures for all four candidates. Model seed 0 and scene index 0 were fixed independently of results. The display is illustrative, not an estimate of event frequency.

For acceptance A, reference cost c_π of the chosen action, and availability $O = \mathbf{1}\{\min_a c_{\mathrm{ref},a} \le \tau\}$, report

$$C = \mathbb{P}(A), \qquad R = \frac{\mathbb{P}(A, c_\pi > \tau)}{\mathbb{P}(A)}, \qquad \eta = \frac{\mathbb{P}(A, c_\pi \le \tau)}{\mathbb{P}(O = 1)}. \tag{2}$$

R is undefined for zero acceptance. The opportunity denominator contains only the supplied four candidates; unsafe accepts never increase η. Scene-level safe loss is $\mathbb{P}(O = 1) - \mathbb{P}(A, c_\pi \le \tau)$. A 2.70 percentage-point loss over all scenes is different from a 3.39% loss of available safe opportunities.

3 What the uniform-cell interpretation changes

Partition the fine grid into coarse cells B with area weights w_B. For one binary property P and footprint F, let $p_B = \overline{P}_B$ and $f_B = \overline{F}_B$, and let $D = \sum_B w_B f_B > 0$. The direct and separately pooled exposures are

$$e_{\mathrm{ref}} = \frac{\sum_B w_B \overline{PF}_B}{D}, \qquad e_{\mathrm{pool}} = \frac{\sum_B w_B p_B f_B}{D}. \tag{3}$$

Subtracting gives the elementary identity

$$e_{\mathrm{pool}} - e_{\mathrm{ref}} = -\frac{\sum_B w_B \operatorname{Cov}_B(P, F)}{D}. \tag{4}$$

This covariance describes spatial coincidence within a cell, not correlations between training scenes. A half-occupied property and half-occupied footprint give exposure .5 from their means, whether their true overlap gives exposure zero or one. That example illustrates the operator; it is neither an empirical frequency estimate nor an impossibility proof for our restricted shape family.

Uniform lifting clarifies the information claim. Let Π replace a field by its mean in every coarse cell. Since ΠP is constant there,

$$\int (\Pi P)(\Pi F) = \int (\Pi P) F. \tag{5}$$

Thus separate pooling is equivalent to uniform lifting of the coarse property followed by integration over the original footprint. A more detailed footprint or constant upsampling alone cannot repair the measurement. An alternative decoder could still use neighboring fractions to infer within-cell geometry. The established claim is therefore residual decision error of the *current target–aggregation interface*, not an irreducible lower bound for every method that reads coarse attributes.

Four levels under one footprint. We compare direct 512 exposure, separately pooled 512 exposure, the actual source 64-to 16 target, and the learned field, always with the same F512 or its exact nonoverlapping 32×32 block means. For a fixed candidate,

$$\widehat{e} - e_{\mathrm{ref}} = (\widehat{e} - e_{\mathrm{target64}}) + (e_{\mathrm{target64}} - e_{\mathrm{pool512}}) + (e_{\mathrm{pool512}} - e_{\mathrm{ref}}). \tag{6}$$

These terms isolate supervision-fit error, source-raster difference, and aggregation error. We reapply Eq. 1 at every level before telescoping *cost* differences. In particular, summing the two channel exposure errors does not give the nonlinear C cost error. Terms may oppose one another; their absolute magnitudes cannot be interpreted as additive shares of final error.

We distinguish two analyses. *Fixed selection* evaluates all levels at the original model-selected action, isolating measurement changes from action changes. *Each-level reselection* lets each level choose once and accept, then evaluates the selected action against the same direct reference. The latter measures full decision consequences, including new failures outside the cases that the uniform oracle originally lost.

Operating conditions and numerical scope. The covariance is zero when either binary field is constant within every cell. For fractional occupancies, Cauchy–Schwarz yields the per-channel envelope $D^{-1}\sum_B w_B\sqrt{p_B(1-p_B)f_B(1-f_B)}$; the sum of channel envelopes also bounds C's cost change because its partial derivatives lie in $[0,1]$. We use joint fractional occupancy and pooled cost margin $|c_{\mathrm{pool}} - \tau| \leq .005$ as fixed diagnostic conditions. The envelope is an algebraic sensitivity bound, not a new deployable safety algorithm. Near-threshold fragility itself is not novel; the empirical question is its frequency and size in the stated generator.

Our precision check rasterizes properties at 1024, averages them to 512, and keeps F512 fixed. It isolates property-raster sensitivity; it is not an exact continuum or footprint-accuracy guarantee. Earlier full 256/direct 512 comparisons are retained separately and are not attribution arms.

4 Decision effects and one prospective confirmation

Consumed discovery. We use the already consumed E3 IID and new-appearance banks, 2,000 scenes each, and all five fixed independent/source 64 models. Original image versions, cards, ties, and stored predictions are retained. The analysis does not recertify any rule. At fixed restored-model actions, pooled 512 versus direct 512 changes 3.68% of labels on IID and 3.83% on new appearance, all reference-safe to pool-danger in these observations. Descriptive crossed model/scene 95% intervals are [2.89%,4.53%] and [3.03%,4.70%]. The five models share 2,000 scenes; their 10,000 model–scene entries are not independent trials.

With independent selection by each oracle, pooled 512 loses 72/2,000 IID safe opportunities (3.60 pp) and 77/2,000 appearance opportunities (3.85 pp). The actual target 64 oracle loses 60 and 67 respectively. These are full-scene consequences, not only differences on unselected high-cost candidates. The fixed-footprint precision check changes .05% of fixed-action labels on IID and none on appearance. Pooling flips stable to that check remain 3.63% and 3.83%.

Signed cost terms at fixed IID selections are+.001807 from pooling, $-.000653$ from the source raster, and $-.000292$ from the model. On appearance they are .002060, $-.000714$, and $-.001453$, giving total mean $-.000108$ despite mean absolute error .003102. Model and net measurement terms oppose one another in 31.52% and 32.49% of decisions. These are signed cancellations, not causal percentages attributed by summing absolute components.

Frozen prediction. This substantive signal motivates one new normal-IID bank of 2,000 scenes, with unchanged models, generator, footprints, formulas, threshold, and ties. No new training is performed. Anonymous continuous patch and full-scene identities have no exact overlap with the

Table 1: Prospective normal-IID confirmation. H1 is the original primary stable pooled 512 endpoint; the target 64 result is supplementary to the full decomposition. One-sided lower bounds use $\gamma = .05/3$; H3's bootstrap is approximate.

Prediction	Observation	Lower bound	Criterion
H1: stable lost opportunities	61/2,000	2.29%	$> 1\%$
H2: only conservative changes in a changed scene	68/68	94.16%	$> 90\%$
H3: small-minus-larger margin enrichment	37.31 pp	26.06 pp	> 10 pp

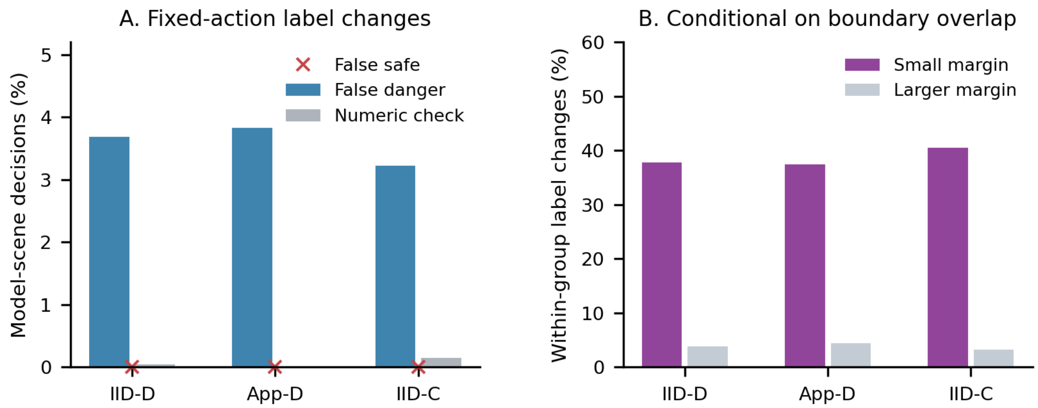

Figure 2: Fixed-action label changes and their operating conditions. IID-D and App-D are consumed discovery banks; IID-C is the prospective IID bank. A: false danger means reference-safe to pooled-danger; false safe is the reverse (no such flips observed). The numeric check holds the footprint fixed. B: rates within overlapping-boundary groups, using the frozen .005 margin. These are observed proportions; Table 1 supplies the separate one-sided hypothesis comparisons.

10,200 E2/E3 records, including training and validation. This is a same-generator confirmation, not cross-task transfer. The three predictions require: (H1) stable-reference lost-opportunity probability above .01 for pooled 512; (H2) conservative direction proportion above .90; and (H3) a small-minus-larger margin flip-rate difference above .10 under joint boundary overlap. Each claim has one-sided error budget $\gamma = .05/3$.

H1 uses an exact binomial lower bound. H3 uses an approximate crossed model/scene bootstrap with only five model seeds. An additional stricter H2 scene-level check was recorded after job submission but before inspecting any new outcome: among scenes with any selected-action pooling flip across the five fixed models, all flips in a scene must be conservative. It avoids a degenerate bootstrap lower limit of one when every observed flip has the same direction. The original protocol and output remain unchanged; Appendix E gives the timing and distinguishes the two estimands.

What changes in decisions. The direct oracle has 1,592 safe opportunities. Pooled 512 accepts 1,529 with no unsafe outcome observed, losing 63 opportunities (3.15 pp of all scenes). The finer-property check also retains a safe candidate in 61 of those scenes: H1's exact lower is 2.2869%. The actual target 64 oracle accepts 1,540, of which two are unsafe, so it loses 54/2,000 safe opportunities. This is 2.70 pp of all scenes and 3.39% of the 1,592 available opportunities. It is not a 2.7% increase in danger, nor a replacement of H1 after seeing results.

At the five models' fixed actions, pooling changes 322/10,000 labels (3.22%, descriptive 95% interval [2.47%,4.02%]), all conservatively. Of these, 312 remain when the finer-property check agrees with the reference label. The check itself changes 15 labels (.15%). Across all nontrivial candidate entries, its p 95 cost difference is .000509, passing the fixed numerical screen. No observed zero-risk count constitutes a new safety certificate.

Table 2: Appended full-scene comparison relative to uniform interpretation. Target counts use 2,000 scenes. Learned counts aggregate five models sharing those scenes and are not 10,000 independent trials. "New unsafe" counts newly unsafe accepted decisions; no previously unsafe accepts are resolved on this bank.

Input	Interpretation	Recovered	New safe loss	New unsafe	R	η
Target 64	Uniform	0	0	0	.13%	96.61%
	Global shift	6	0	3	.32%	96.98%
	PLIC	44	0	16	1.13%	99.37%
Five models	Uniform	0	0	0	1.03%	96.77%
	Global shift	0	0	0	1.03%	96.77%
	PLIC	83	0	43	1.56%	97.81%

Among overlapping-boundary actions, the frozen small-margin group has 40.57% label changes versus 3.26% at larger margins. All 68 scenes with any flip across models have only conservative flips, with exact conditional lower 94.1566% for the fixed model bank. The evidence confirms the predicted operating conditions in the normal distribution. It does not make near-threshold sensitivity a new general phenomenon, prove all pooling errors are positive, or establish that every coarse decoder must lose these opportunities.

5 One same-information reconstruction control

The preceding replacement fixes learning error but retains the aggregator. We now test whether a different interpretation can exploit information already present in neighboring coarse values. Development uses the consumed 2,000-scene IID discovery bank. One implementation and two scalar offsets are fixed before evaluating the already consumed 2,000-scene confirmation bank. This is an *appended baseline analysis*, not a new untouched confirmation. There is no new training, dataset, resolution search, or risk certification.

Three treatments with identical action access. The primary track uses the perfect actual target 64 field; the secondary track uses all five existing restored model fields. Uniform interpretation computes the original exposure. All three treatments may access exactly the same F512, so PLIC is not given better action geometry. The global shift subtracts one constant after Eq. 1. Its argmin is the original uniform argmin, since translation preserves ranking; no clipping-induced ties are introduced. From a fixed 13-value offset bank, development maximizes safe-accepted count subject to no increase in observed unsafe-accepted count relative to uniform. Ties favor the smallest absolute offset. The selected offset is .001 for the target track and zero shared across the five learned models.

The only spatial alternative is the established Parker–Youngs PLIC construction (Pilliod & Puckett, 2004). A fixed weighted 3×3 gradient with center weight 2 estimates a cell's outward normal. A straight half-plane then matches the cell's original fraction by analytic square-area inversion. Empty/full cells remain unchanged; cells with gradient L1 norm at most 10^{-12} retain uniform occupancy rather than an invented orientation. Exact line-cut fractions of fine pixel squares are integrated against F512. Coarse predictions are neither thresholded nor denoised. The reconstruction routine accepts only the coarse field and footprints, never true patch parameters, fine property masks, reference costs, or oracle boundary normals. It is a first-order PLIC baseline, not the second-order ELVIRA method or a newly proposed algorithm.

Full-scene gains and harms. Table 2 reports every scene, including no-safe-candidate cases, and both recovered and newly harmed decisions. PLIC recovers 44 of the target oracle's 54 lost opportunities without newly losing a previously accepted safe one. However, it introduces 16 unsafe accepts: total unsafe count rises from 2 to 18. Safe opportunity efficiency rises from 96.61% to 99.37%, while empirical unsafe frequency among accepted rises from .13% to 1.13%. The frozen shift recovers six opportunities and introduces three unsafe accepts.

The learned track also has a mixed outcome. PLIC recovers 83 safe model–scene decisions and adds 43 unsafe accepts, with no new safe loss observed. Net safe acceptance increases .83 pp (descriptive crossed 95% interval [.51,1.19] pp), while new unsafe acceptance increases .43 pp ([.20,.69] pp).

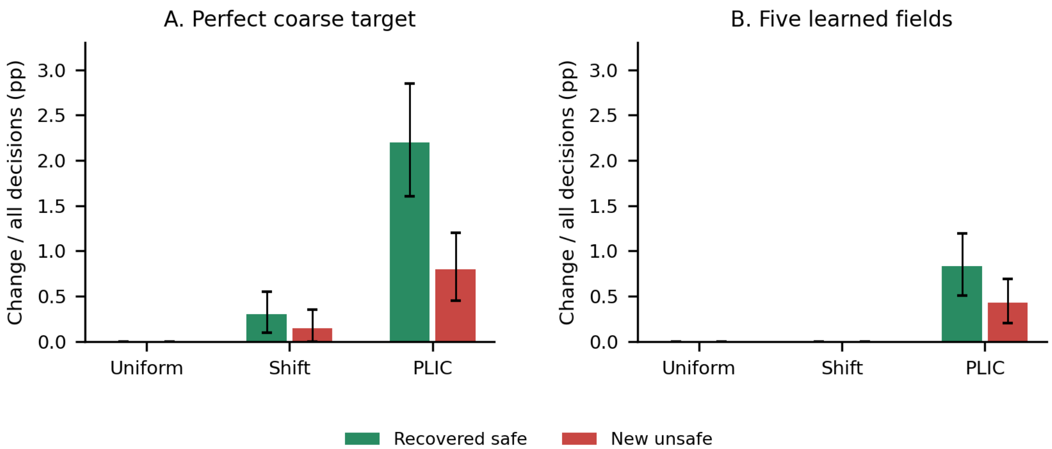

Figure 3: Opportunity recovery and newly unsafe acceptance must be read together. PLIC uses the same coarse field and action information as uniform integration. No new safe losses were observed, but both input tracks add unsafe accepts. The offset was fitted under a development unsafe-count constraint that was not imposed on PLIC; these selected points are not a matched-risk comparison.

Table 3: First event satisfying the stated cancellation pattern, in scene/model order. The same reference-unsafe action is rejected before reconstruction and accepted after it ($\tau = .02$). This is a conditional illustration; all 43 newly unsafe learned events remain in the archive.

Quantity on the same action	Uniform	PLIC
Reference cost	+0.024972	+0.024972
Target estimate	+0.031961	+0.021431
Learned estimate	+0.021279	+0.016964
Target-measurement cost error	+0.006989	-0.003541
Model-relative-to-target cost error	-0.010682	-0.004467
Total cost error	-0.003693	-0.008008
Decision	Reject	Accept (unsafe)

The target track's net safe gain is 2.20 pp ([1.60,2.85] pp). The selected learned shift is zero and changes no outcome. Fixed-selection analysis yields the same recovery and new-unsafe totals on this appended bank, so those totals are not explained solely by permission to reselect; the event archive retains action identities and individual changes.

A concrete cancellation failure. For 25 of the 43 newly unsafe learned decisions, the action stays unchanged, baseline model and target-measurement errors have opposite signs, target-measurement absolute error decreases under PLIC, and total absolute error increases. These conditions are checked on each event, not inferred from aggregate means. For the first such event in scene/model order, Table 3 shows that both component absolute cost errors improve, yet their earlier opposition disappears and the total underestimation worsens. All 43 events, including the 18 not meeting this joint pattern, are retained. The example supports a particular failure mechanism, not a claim that cancellation explains every harmful change.

What this comparison establishes. The recovery demonstrates that the current coarse field contains spatial structure usable by another interpretation; the original residual loss cannot simply be called irreducible representation error. It does not demonstrate a risk-free repair or universal superiority over scalar calibration. The chosen offset is constrained not to add development unsafe outcomes, whereas PLIC is not. The full development shift curve is therefore disclosed in Appendix C: larger offsets recover more opportunities at higher risk. We do not tune another offset on the appended bank. The conclusion is to separate supervision fidelity, spatial interpretation, and acceptance calibration,

Table 4: Supporting results from distinct earlier banks. E3's fixed-restored rule was already certified in its original finite bank. Those certificates do not transfer to the appended shift or PLIC rules. R is unsafe among accepted.

Study/bank	Rule	Acceptance	R	η
E1 appearance	Inherited -2	85.03%	4.70%	99.63%
E1 appearance	Same weights, restored	80.97%	1.73%	97.83%
E3 IID	Restored, fixed .02	78.58%	1.07%	96.93%
E3 IID	Selected certified threshold	81.29%	2.51%	98.82%
E3 IID	Joint residual protection	0%	Undefined	0%

and to re-evaluate the latter after changing the former. No earlier certificate is inherited by either modified rule.

6 Supporting controls and relation to earlier work

Learning and decision-rule effects remain distinct. A prior matched geometry $\times$ source-raster experiment trained 20 models at equal budgets. Independent geometry improves new-appearance regret relative to the balanced-anchor scheme at both source 48 and source 64, with crossed 95% intervals [.003101,.006853] and[.002850,.017690], each exceeding the predeclared .001 effect threshold. This controls a genuine learning-related effect while keeping common-reference scoring fixed. The intervention changes shape, size, rotation, and position jointly; it does not isolate position alone.

Earlier fixed-weight and risk-control results are summarized in Table 4; details are in Appendix F. They support the argument without defining its novelty. E1 uses historical checkpoints and consumed diagnostic data. E3 uses the five E2 independent/source 64 models with 1,600 new calibration and 2,000 test scenes per domain. Their residual and certification results must not be treated as one within-model factorial test.

Joint protection and selective-risk control protect different objects. If nonnegative estimates have a joint residual quantile $q > \tau$, adding that bound necessarily rejects all actions. A rule controlling risk among accepted actions need not protect every unselected candidate. The E3 exact-binomial finite-bank procedure applies established Learn then Test principles (Angelopoulos et al., 2021); its success shows that an imperfect pipeline can be useful under a specified rule and distribution. It neither validates its measurement interface nor supplies certificates for modified aggregation.

Spatial reconstruction and task-specific representations. Volume-of-fluid reconstruction estimates an interface from grid-cell fractions. Pilliod & Puckett (2004) compare first- and second-order methods, including the Parker–Youngs gradient reconstruction used here. That literature supplies both a baseline and the reason not to equate uniform-cell failure with representation impossibility. Our task has noisy learned fractions, swept action queries and thresholded decisions, rather than interface advection; we do not introduce a new reconstruction scheme or claim second-order accuracy for this baseline.

Lambda-Field studies the compatibility of occupancy representations and path-risk integration (Laconte et al., 2019). Its collision-risk construction differs from our public mean-exposure cost, but representation-dependent path risk is established prior work. Task-based quantization likewise studies limited representations in terms of a downstream objective rather than signal recovery alone (Shlezinger et al., 2019). Smart Predict, then Optimize explicitly separates prediction and decision losses (Elmachtoub & Grigas, 2022). Our narrower contribution is a same-reference empirical attribution with actual acceptance effects, prospective operating-condition evidence, and a same-information alternative interpretation.

Egocentric Conformal Prediction already addresses conservatism from safety- irrelevant prediction errors (Shin et al., 2025). Conformal Decision Theory calibrates decision loss directly (Lekeufack et al., 2024). Neither decision relevance nor moving beyond prediction-set calibration is a novelty claim here. We use these distinctions to keep measurement validity and the object of a statistical guarantee separate.

7 LIMITATIONS AND CONCLUSION

The task is synthetic, uses a known cost law, and restricts actions to four candidates. Numerical checks condition on a fixed footprint and do not prove continuum accuracy. Oracle boundary features are privileged diagnostics. The prospective confirmation concerns the original mechanism; the added reconstruction and shift are retrospective comparisons on consumed data. Five model seeds limit training-distribution inference. A single first-order PLIC baseline cannot characterize every same-information decoder, and successful oracle-input repair would not by itself establish safe learned-model deployment.

An attempted real-image extension using RELLIS-3D (Jiang et al., 2021) stopped before mechanism evaluation because the geometric reference lacked independent error bounds. Its initial 24-frame screen and subsequent prerequisite stop are retained in the appendix; the planned new 48-frame set was never instantiated. It supplies neither supporting nor refuting evidence for cross-task transfer. The extension remains closed; no VLM, RL or external experiment is added.

Accurately realizing a coarse supervision target does not, by itself, validate the action measurement implemented by its current aggregation rule. In this controlled task, common-reference oracle replacements isolate a primarily conservative aggregation effect with material opportunity loss and predictable operating conditions. Distinguishing spatial interpretation from the underlying information avoids an unsupported impossibility claim. The appropriate technical choice is determined by what is recovered and what new errors are introduced, not by component fit or recovered failures alone.

AI USE STATEMENT

Generative AI tools assisted with research questions and hypotheses, experimental design, procedural-data and analysis implementation, mathematical exposition, interpretation, literature discovery, figures, and manuscript drafting and editing. Numeric results were produced by executable analysis, not inferred from generated prose. Independent computational checks are documented in the appendix. This is a working draft: human authors must review the AI-assisted contributions and finalize responsibility statements before submission; completed human review is not asserted here.

REPRODUCIBILITY STATEMENT

The appendix specifies data provenance, numerical definitions, fixed training, the retrospective status of the appended comparison, and verification procedures. Source freezes, results, and earlier negative outcomes are preserved internally. The present manuscript source contains no personal paths or development-repository links; an anonymous code supplement must still be finalized before submission.

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A Data separation and inference units

E2 crosses two geometry schemes, source rasters 48/64 and five paired seeds (20 models), with 800 training and 100 validation scenes per condition. Training follows Section 2; the source rasters share RGB and the final checkpoint is fixed without validation selection. Each independent-geometry test bank has 400 scenes. Training uses one assignment version per scene; evaluation retains both but samples one version/card for the actual decision.

E3 preselects all five independent/source 64 E2 checkpoints. It uses 400 development scenes and, per domain, 1,600 calibration and 2,000 test scenes. The normal generator and independent path sampler in Section 2 operate before rasterization; no pixel, risk or feasibility screen changes their sampling.

Oracle discovery reuses the consumed E3 IID/appearance test banks. The later prospective bank has 2,000 new IID scenes (geometry seed 2612091301), with a recorded source/checkpoint freeze before generation. Exact anonymous patch and complete- scene identities are disjoint from all 10,200 E2/E3 records. IDs, appearance names, seeds and path/patch ordering are excluded from the identity comparison; arbitrary geometric symmetries are not quotient out. After that confirmation, the bank is consumed. The PLIC/shift analysis in Section 5 is explicitly appended to it, without new samples or renewed confirmatory status.

Learned-track ratios pool five models over the same 2,000 scenes; crossed bootstrap draws preserve that dependence. Target-oracle intervals resample scenes only. Five training seeds still limit the approximate crossed inference.

B Spatial reference and reconstruction details

The direct reference rasterizes each continuous property and swept-union footprint at 512. It averages their product over footprint area, then applies the public card law. It is a fixed discrete reference for synthetic mean exposure. The auxiliary property check rasterizes properties at 1024, averages to 512 and retains F512. An earlier full 256/direct 512 comparison changes both property and footprint rasters and is kept separate from the common-footprint attribution.

Signed cost attribution. Exposure replacement follows Eq. 6. For every fixed action, cost differences are separately evaluated as

$$\widehat{c} - c_{\text{ref}} = (\widehat{c} - c_{\text{target}}) + (c_{\text{target}} - c_{\text{pool}}) + (c_{\text{pool}} - c_{\text{ref}}).$$

The nonlinear formula is applied before differencing. Channel exposures, signed/absolute costs and label-pattern transitions are preserved. The variance envelope uses privileged reference occupancies; its zero-flip implications are implementation checks.

PLIC area inversion. The normal is the negative weighted centered gradient of the coarse fraction, with transverse weights $(1, 2, 1)$ and edge-replicated padding. Pairing differences before summation eliminates roundoff-generated transverse components for an exactly constant transverse field. Normalize (n_x, n_y) by its L1 norm. For a nonzero normal let $a = |n_x|$, $b = |n_y|$, $a + b = 1$, $s = \min(a, b)$ and $l = \max(a, b)$. The area CDF of $aU + bV$ for independent unit-square coordinates can be evaluated symmetrically. At $0 \le t \le 1/2$ its lower half is

$$H(t) = \begin{cases} t^2/(2ab), & t < s, \\ (t - s/2)/l, & t \ge s. \end{cases}$$

Axis-aligned limits are handled analytically; the upper half uses $1 - H(1 - t)$. For fraction p, invert the lower tail $q = \min(p, 1 - p)$ using $t = \sqrt{2abq}$ when $q \le s/(2l)$, otherwise $t = lq + s/2$; reflect for $p > 1/2$ and translate for negative normal components. This yields a half-plane whose unit-square area is the original p.

For each fine pixel square, the same CDF gives its exact fractional area under the reconstructed half-plane. Integrating those fractions against binary F512 preserves each coarse cell's original mass. Cells where a footprint is entirely zero or one need no fine calculation: the integral is fixed by conserved mass. All fractions, including small learned probabilities, are retained. A mixed cell whose gradient L1 norm is at most 10^{-12} stays uniform. This is an explicit fallback, not a inferred true boundary.

On the appended bank, the maximum analytic inverse-area discrepancy is 2.78×10^{-16} and the maximum sum-of-fine-areas discrepancy is 3.33×10^{-16}, below the fixed 10^{-10} tolerance. There are 34,837 gradient-fallback cells across the input tracks and 5,047,140 reconstructed query cells. These numerical checks establish conservation, not physical correctness of the inferred boundary or a safety guarantee.

C Shift fitting and unequal risk points

The fixed cost-offset bank is

$$\{-.02, -.015, -.01, -.005, -.002, -.001, 0, .001, .002, .005, .01, .015, .02\}.$$

One offset is fitted for perfect target input and one is shared across the five model inputs. Fitting uses only consumed IID discovery data: maximize safe acceptance subject to unsafe-accepted count no larger than uniform. Ties prefer smallest absolute offset and then smaller offset. The rule is frozen at .001 for target input and 0 for learned input. It is empirical fitting, not a risk certificate or guarantee that the same unsafe-count constraint will hold later.

The entire development bank is retained. At target offset .005, development has 1,578 safe and 17 unsafe accepts; at .01 it has 1,598 safe and 33 unsafe accepts. Fixed PLIC has 1,600 safe and 16 unsafe accepts on development. The selected .001 shift has 1,550 safe and one unsafe accept. Larger shifts therefore can recover more opportunities, and the selected point must not be interpreted as the capacity of the whole shift family. No alternative offset is chosen using the appended outcomes. The different risk levels preclude claiming a matched-risk deployment advantage from the reported selected points.

Table 5: Absolute appended outcomes. Learned rows pool five models sharing 2,000 scenes. The direct oracle has 1,592 safe opportunities per model.

Input	Interpretation	Accepted	Safe accepted	Unsafe accepted	Safe opportunities lost
Target	Uniform	1540	1538	2	54
	Shift	1549	1544	5	48
	PLIC	1600	1582	18	10
Learned	Uniform	7783	7703	80	257
	Shift	7783	7703	80	257
	PLIC	7909	7786	123	174

D Event-level cancellation evidence

Each recovery, new safe loss, new unsafe accept and resolved unsafe accept keeps its scene/model/card/action identity. At the original uniform-selected action, write target-measurement error as $m = c_{\text{target}} - c_{\text{ref}}$ and model-relative error as $l = \widehat{c} - c_{\text{target}}$. After changing interpretation, retain both m' and l', with $\widehat{c}' - \widehat{c} = (m' - m) + (l' - l)$. This fixed-action equality is checked before interpreting reselected outcomes.

For the 43 newly unsafe learned PLIC decisions, 25 satisfy all of: unchanged action, $ml < 0$ before modification, $|m'| < |m|$, and $|m' + l'| > |m + l|$. The other 18 are retained as counterevidence to an exclusive cancellation explanation. The first matching event in scene/model order is scene 264, model index 1, exchanged version, card D, action 0. In it, both $|m|$ and $|l|$ decrease yet total underestimation grows, as detailed in the main text. This is a transparent conditional illustration; it was not chosen to estimate the population frequency of such cases.

E Prospective direction check and its addendum

The original confirmation uses one-sided budgets .05/3. H1 counts scenes where direct 512 and the finer-property check both retain at least one safe candidate, but pooled 512 fails to produce a safe accepted action. Its 61/2,000 events give exact lower .02286937. H2 initially uses a crossed bootstrap for the fraction of fixed-selected pooling flips that are conservative. Since all 322 flips are conservative, its bootstrap lower degenerates to 1 and cannot establish population certainty. H3 compares small/larger margin flip rates under joint boundary overlap; its estimate is .373080 and approximate lower .260586.

The direction addendum was recorded at 20:57:54 UTC after job submission but before any fresh outcome was inspected and before the original hypothesis file was generated at 21:00:24 UTC. It is disclosed as a strengthening after the original freeze, not silently included in that freeze. It requires an additional exact scene-level check at the same H2 error budget: conditional on a scene having any flip across the fixed five-model bank, every flip must be conservative. All 68 changed scenes satisfy it; the exact lower is .941566. This conditional fixed-bank estimand differs from the model–scene weighted proportion. Both original outputs and the added check are preserved.

F Earlier controls and closed external attempt

E1 uses five historical models and a consumed 300-scene calibration bank. With restored output, positive-residual quantiles are .02797–.03287 for all 24 pair/version/card/candidate entries, .02201–.02767 for the current four candidates, and .00501–.00790 for the fixed selected action. These are three different protection objects. The smaller marginal selected-action bound does not itself control unsafe frequency after acceptance.

E3 separately uses all five preselected E2 independent/source 64 models. Twelve threshold rules per source/target calibration domain use exact one-sided Clopper–Pearson upper bounds with family budget $\delta/(5 \cdot 2 \cdot 12) = .05/120$. The desired conditional unsafe limit is $\epsilon = .05$, distinct from danger threshold $\tau = .02$ and residual miscoverage $\alpha = .05$. The original fixed .02 restored rules already certify, with source upper bounds 1.66–2.54%; selecting a higher-coverage certified threshold increases both opportunity use and test risk. Source certificates do not become unknown- distribution guarantees on appearance shift, and no old certificate applies to PLIC or a newly fitted shift.

The RELLIS attempt evaluated a fixed 24-frame training-sequence pilot. All-four corridor unknown area at most 5% held in 0/24 cases; an optimistic diagnostic discarding depth/occlusion checks reached 3/24. These results concern the chosen single-scan surface-fit interface, not the dataset's general suitability. A bounded revision inspected 240 raw scans around those same 24 development frames and ran a motion-compensation prototype. Independent physical alignment and visibility error bounds remained unresolved. It stopped before selecting, acquiring or evaluating the planned new 48-frame set. This is a prerequisite stop, not 0/48 and not a mechanism-transfer result; neither external training nor an external final test was run.

G Computational verification

The original oracle round has six distinct tests and independent replay of 52,000 discovery decisions and 26,000 prospective decisions, with 768 and 384 blockwise candidate-cost checks. The additional PLIC tests cover mass against independent polygon clipping, uniform-lift identity, zero-gradient fallback, complement/transposition behavior, axis orientation, and constant-footprint cells. One initial test detected transverse roundoff at approximately 8.3×10^{-16} for an exactly constant transverse field; changing the arithmetic to pair differences before summing fixed it. The original test, failed attempt, and unchanged scientific tolerance are retained; no development scene was read before the repaired tests passed.

Independent appended verification replays 72,000 decisions and 65,536 fine-square areas with polygon clipping. The maximum polygon difference is 3.33×10^{-16}; the constant-lift identity differs by at most 3.71×10^{-14}. It also verifies all 25 specified cancellation-pattern events among the 43 new unsafe learned accepts. All computation runs on the cluster; frozen inputs and outputs are preserved.